LLM4Rec Final Exam Report — Repeat Viewing Dominates: TV Programme Recommendation on sweet.tv

Student: Anastasia Shavrina (Шаврина Анастасия Олеговна)
Team: Individual
Email: shavrina157@gmail.com
Date: 2026-10-30
Assignment: Final exam quiz (FKN) — TV programme recommendation on sweet.tv (Kaggle account: Maizawitch)
Code: https://github.com/shavrinaanastasia/sweettv-recsys-exam

Abstract

We predict, for 1,260 sweet.tv users, the five TV programmes they will watch most often in weeks 31-42, given a channel-level view log (weeks 11-30) in which only about 11% of schedule rows carry a programme id. We join views to labelled airings with the host's own "≥ 80% watched" rule, compare a popularity baseline with implicit ALS, content-based filtering and personalized PageRank on one temporal hold-out, and combine them by score fusion followed by a LightGBM LambdaRank re-ranker. The final strategy reaches MAP@5 = 0.352 on untouched validation users (popularity: 0.224) and a returned Kaggle private score of 0.43344 (popularity: 0.28773). The main lesson is that this target rewards predicting repeat viewing on a user's habitual channels; content similarity alone does not even beat popularity.

Index Terms — implicit feedback, personalized PageRank, matrix factorization, learning to rank, TV recommendation

1. Introduction

Problem.
sweet.tv logs which channel each user watched and when, and publishes the broadcast grid separately. For each of 1,260 test users we must output five programme ids, ordered, that best match the five programmes the user will watch most often (≥ 80% of an airing) in the next 12 weeks; the metric is MAP@5 [1].

Motivation.
A TV service's home screen has room for a handful of tiles. If they reflect what a viewer will really come back to (the evening news, a children's series) the screen saves effort; if not, it is ignored. The data also forces two non-trivial steps: views must first be turned into user-programme interactions, and every method must extrapolate forward in time.

Concrete example.
User 8377619604347126107 watched 179 airings to ≥ 80% in weeks 11-20, spread over 22 programmes. In weeks 21-30 their five most-watched programmes were 'ТСН. Неделя', 'Сегодня. Итоги', 'Факты. День', 'Завтрак с 1+1' and a 'Свинка Пеппа' series id — all five already in their history. Personalized PageRank returns exactly these five (AP@5 = 1.0, Table 4).

Contributions.
(i) A reproducible join that applies the host's label recipe, and a validation set built with the same recipe. (ii) A like-for-like comparison of four methods plus a channel-exposure signal, and a two-stage hybrid whose gain over every single method holds both on held-out validation users and on the Kaggle private split.

2. Related Work

Prior work consulted:

- **Implicit-feedback matrix factorization** [2] turns counts into binary preference with a confidence weight and solves it by alternating least squares; our M1 implements those update equations directly.
- **Personalized / topic-sensitive PageRank** [3] biases the random-walk restart vector towards a chosen set of nodes; our M3 restarts on each user's own history over a programme co-view graph.
- **Content-based recommendation** [4] builds item vectors from attributes and a user profile from liked items; our M2 uses TF-IDF over category, genres, decade, director and actor tokens.
- **Learning to rank**: LambdaMART [7] optimises a ranking metric through gradient-boosted trees; we use the LightGBM implementation [6] for the re-ranking stage.
- The competition page [1] defines the data, the metric and the exact label recipe.

Alternatives considered. Association rules [5] were not implemented: with 6,000+ programmes and per-user baskets dominated by daily repeats, rules mostly restate "watches A → watches A again", which history already captures. Sequence models (session RNN / transformers) were not tried: views are channel intervals, not item clicks, and only 20 weeks of history exist. A two-tower model was not tried for the same reason and because the hand-built graph and factorization already give strong collaborative signals at a fraction of the cost.

3. Method

Approach.

Turn the view log into (user, programme, number of airings watched ≥ 80%) interactions, score every candidate programme with each method, and rank. Every method shares one candidate pool, one fallback and one evaluation protocol. The final system uses the fixed-weight score fusion of all signals as a candidate generator (top 150 per user) and re-orders those candidates with a LambdaRank model.

Pipeline.

- *Join.* A view (user, channel, [start, stop]) is matched to a schedule row on the same channel whose airing [start, start + duration] overlaps it. Overlap seconds of all of the user's views are summed per airing and capped at the airing duration. *Watched* means overlap / duration ≥ 0.8 — exactly the host's label rule [1]. Figure 1 justifies this cut on the data as well: screen time is U-shaped (zapping near 0, full watches at 1), and 0.8 sits in the flat valley, so small changes to it move few pairs. 20.4% of 3,014,129 user-airing pairs pass.
- `tv_show_id = 0`. Unlabelled airtime is never an item (the host removed it from the labels). It is used only as context: each user's viewing seconds per channel, over **all** airtime, feed the channel-exposure signal below.
- *Labels* (validation): per user, count airings watched ≥ 0.8 per programme in the target weeks, keep users with ≥ 5 such programmes, take the 5 most frequent (ties: more watched seconds).
- *Candidate pool:* labelled programmes that air in the target weeks — for the test, read from the weeks 31-42 *schedule* only. Re-recommending a programme the user already watched is **allowed for every method** (the target is repeat-heavy, and the host does not forbid it).
- *Fallback:* empty slots are filled from the popularity ranking, identically for every method. Each method turns its scores into a list by sorting its own scores inside the pool; scores are never compared across methods without per-user min-max normalisation.

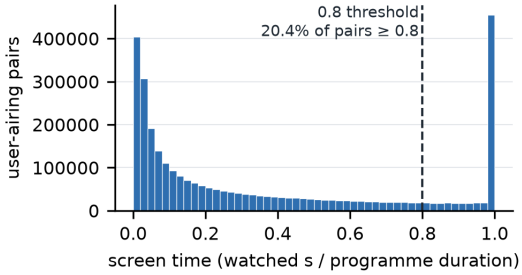


Figure 1: Screen time of all 3,014,129 user-airing pairs (validation and test history). The 0.8 cut is the host's label rule.

Methods.

- *Popularity (baseline):* recency-weighted number of airings of the programme watched ≥ 0.8 by all users in the train weeks (half-life 4 weeks).
- *History (reference, not a counted method):* the user's own count of ≥ 0.8 airings per programme.
- *M1 implicit ALS* [2]: r = recency-decayed counts, confidence c = 1 + α r; 16 factors, λ = 30, α = 1, 10 iterations, seed 42.
- *M2 content-based* [4]: TF-IDF item vectors over 17,193 metadata tokens; user profile = count-weighted sum of watched items; cosine score.
- *M3 personalized PageRank* [3]: item graph with edge weight = number of users who watched both items ≥ 20 min; row-normalised; restart probability 0.3 to the user's normalised history; 20 power iterations.
- *Channel exposure (context signal):* Σ over channels of (user's share of viewing seconds on channel c) × (airings of the item on c in the target weeks).
- *Hybrid:* per-user min-max normalised scores, weighted sum; weights by coordinate ascent on validation half A: channel 2.0, PPR 1.5, history 0.75, ALS 0.5, CBF 0.1, popularity 0.05.
- *Final: hybrid + re-ranker:* top 150 fused candidates per user, re-ordered by LightGBM LambdaRank [6], [7] (300 trees, learning rate 0.05, 31 leaves, ≥ 100 rows per leaf, bagging 0.8, seed 42, deterministic) on 31 features: every component's raw score, normalised score and rank, fused score and rank, watch counts and seconds, target/train airing counts and their ratio, user activity. Relevance = 5 for the label's #1 down to 1 for #5.

Key design decisions.

(1) *Use the host's 0.8 rule rather than an absolute-seconds cut,* because the score is computed on labels built with it; an absolute cut would mislabel long films and short news equally. (2) *Fit the re-ranker without validation labels:* for validation it learns history 11-15 → labels 16-20 and is then scored on 11-20 → 21-30; for the test it learns 11-20 → 21-30 (training data at that point) and is applied to 11-30 → 31-42. (3) *Keep PPR's full graph* (dense matrix product) after pruning to the 50 strongest edges per item cost up to 0.05 MAP (§5).

Tools & configuration.

Python 3.11.9; numpy 2.4.6, pandas 3.1.0, scipy 1.17.1, pyarrow 26.0.0, LightGBM 4.7.0 (native `lgb.train` API), matplotlib 3.11.2, reportlab 5.1.0, Kaggle CLI 2.2.5. ALS and PPR are implemented from the equations in [2], [3] (`src/methods.py`), not from a library. Windows 11, 8 logical CPUs, 16 GB RAM, no GPU. Seeds: user A/B split 0, ALS 42, LightGBM 42. Runtime: parse 42 s, join 12 s; scoring the 1,260 test users: PPR 41 s, ALS 6 s, CBF 0.3 s (`results/predict_summary.json`); re-ranker test run 183 s end-to-end.

4. Experiments

Setup.

View log: 5,821,987 views by 4,914 users on 202 channels (weeks 11-30); schedule: 381,892 labelled airings. Only 10.8% of schedule rows (16.0% of airtime) carry an id, yet 52.4% of all viewing seconds fall on that labelled airtime. *Validation split:* train weeks 11-20 → labels from weeks 21-30, giving 4,030 users with five labels (1,150 of them are Kaggle test users). The test target window is later than all training weeks, so this split preserves temporal order and has a similar history-to-horizon shape (10/10 weeks vs. 20/12 for the test). Users are split 50/50 at random (seed 0): every hyper-parameter and fusion weight is chosen on half A (2,048 users); half B (1,982 users) is reported untouched, mirroring the leaderboard's public/private user split [1]. Cold start: all 1,260 test users appear in the training log, so there is no user without any views; 22 of them have fewer than five programmes with ≥ 0.8 history, so their history-only list is topped up from the popularity fallback (`results/predict_summary.json`).

Method	Val. A	Val. B (untouched)	Val. test users	Kaggle public	Kaggle private
Popularity (baseline)	0.2226	0.2244	0.2486	0.30344	0.28773
History (reference)	0.3105	0.3029	0.3468	—	—
M1 implicit ALS	0.2890	0.2862	0.3120	0.37750	0.36917
M2 content-based	0.1871	0.1831	0.2036	0.22711	0.21839
M3 personalized PageRank	0.3239	0.3182	0.3610	0.39959	0.38530
Channel exposure (signal)	0.3227	0.3192	0.3477	—	—
Hybrid, fixed weights	0.3484	0.3420	0.3810	0.44153	0.42921
Hybrid + re-ranker (final)	**0.3563**	**0.3524**	**0.3933**	**0.43978**	**0.43344**

Table 1: MAP@5. Validation = train 11-20, labels 21-30; Kaggle = train 11-30, hidden labels 31-42. Sources: `results/hybrid_val.json`*,* `results/rerank_val_n150.json`*,* `results/kaggle_submissions.csv`*.*

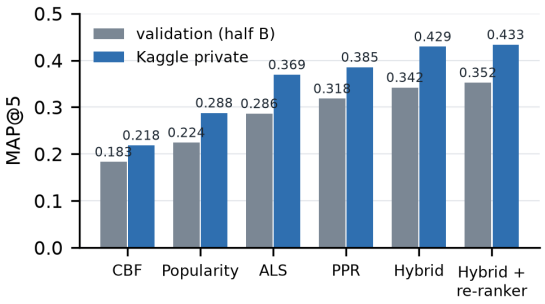


Figure 2: Validation MAP@5 (half B) and returned Kaggle private MAP@5 per method. The order of the methods is the same on both.

Comparison vs. baseline.

Every collaborative signal beats popularity by a wide margin: ALS +0.062 and PPR +0.094 on half B; the final strategy +0.128 on half B and +0.146 on Kaggle private. Content-based filtering is the only method below the baseline (−0.041 on B). The methods rank in the same order on validation and on Kaggle private (Figure 2); Kaggle scores are higher overall, plausibly because test predictions use 20 instead of 10 weeks of history — a hypothesis we did not test.

Hybrid and ablation.

Removing one component from the fixed-weight hybrid (half B, `results/hybrid_val_run1.json`): channel −0.019, PPR −0.004, history −0.002, CBF −0.001, popularity −0.000, ALS +0.000. A second weight search with a wider grid (up to 6.0) ended *lower* on half A (0.3477 vs. 0.3484) and was not used. The re-ranker was compared over two candidate depths and two model sizes on half A (150 candidates, small model best); recall of the label inside the 150 candidates is 0.767, which bounds what re-ranking can recover.

Tuning record.
ALS needed three grids because the best point twice sat on the grid edge; the final λ = 30 beats 10 and 100. Weighting ALS by watched seconds instead of ≥ 0.8 counts collapsed MAP to 0.02-0.06 (plausibly because zapping visits then dominate; not tested further). Every grid point, including the losing ones, is logged in `results/val_runs.log` and `results/val_log_*.json`.

Verification.
(i) MAP@5 code reproduces the course slide's worked example (AP = 0.917 and 0.533). (ii) One user-airing overlap recomputed from the raw CSV with the standard `csv` module equals the join output (5,973 s of a 6,000 s film). (iii) All 4,030 validation label sets contain 5 distinct programmes, each watched ≥ 0.8 in weeks 21-30. (iv) ALS objective decreases every iteration (49,593 → 26,627 over 6 iterations). (v) PPR transition rows sum to 1, and each user's score vector keeps total mass 1. (vi) Each submission file has 1,260 rows, five distinct non-zero ids from the test-week pool, and was scored COMPLETE by Kaggle. (vii) The dense and sparse PPR implementations return the identical score (A = 0.32295) on the same graph. (viii) I computed AP@5 by hand for the heavy viewer's popularity list (Table 4), using AP@5 = (1 / min(5, |R|)) · Σ_k P@k · rel(k): the hits are at ranks 1, 3, 4 and 5 and |R| = 5, so AP@5 = (1/5)(1 + 2/3 + 3/4 + 4/5) = 0.6433, the same value the code reports (`results/examples.json`); my working is shown in Figure 3. Sources: `results/checks.json`, `src/checks.py`.

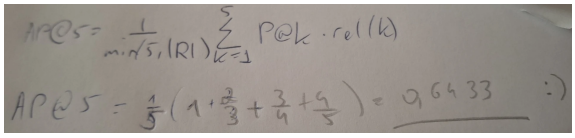


Figure 3: My hand calculation of AP@5 for the heavy viewer's popularity list (photo, cropped).

4.1 Recommendation examples

Users were chosen by a fixed rule, not by inspecting results: among validation users who are also Kaggle test users, the one closest to the 90th percentile of training activity (heavy) and the one closest to the 10th percentile (light). Lists are from the validation models (train 11-20) so they can be compared with what the users actually watched in weeks 21-30. Titles are the schedule text of the programme's latest airing. One id can cover many airings with different titles: 12002955 appears under 207 episode titles of 'Свинка Пеппа', and 2400480 covers both the weekly 'ТСН. Неделя' and the daily 'ТСН.' news. In Tables 4-5, **bold** = in the user's actual weeks 21-30 label; ⁿ = not watched ≥ 80% by this user in weeks 11-20.

Method	#1	#2	#3	#4	#5	AP@5
Actual 21-30	*2400480*	*2400508*	*240081*	*20088*	*12002955*	
Popularity	**2400480**	12001682	**240081**	**20088**	**2400508**	0.643
History (ref.)	**2400480**	**240081**	**12002955**	200338	**2400508**	0.760
M1 ALS	**2400480**	**20088**	**240081**	**12002955**	200344	0.800
M2 CBF	**2400480**	**240081**	2500261ⁿ	5900347ⁿ	2400467	0.400
M3 PPR	**2400480**	**240081**	**12002955**	**2400508**	**20088**	1.000
Hybrid (fixed)	**2400480**	**12002955**	**240081**	**20088**	12002896	0.800
Final: re-ranker	**2400480**	**240081**	**12002955**	**20088**	12002896	0.800

Table 4: Heavy viewer, user 8377619604347126107 (179 airings watched ≥ 80% in weeks 11-20, 22 distinct programmes).

Method	#1	#2	#3	#4	#5	AP@5
Actual 21-30	*20088*	*2400480*	*90077275*	*400455*	*90079862*	
Popularity	**2400480**	12001682ⁿ	240081ⁿ	**20088**	2400508	0.300
History (ref.)	6500479	**2400480**	90072234	**20088**	90057599	0.200
M1 ALS	**20088**	**2400480**	240081ⁿ	2400508	2400467ⁿ	0.400
M2 CBF	**2400480**	5900347ⁿ	2400467ⁿ	240081ⁿ	2500261ⁿ	0.200
M3 PPR	**2400480**	6500479	**20088**	400363	90021962	0.333
Hybrid (fixed)	**2400480**	**20088**	6500479	240081ⁿ	2400508	0.400
Final: re-ranker	**2400480**	**20088**	2400508	240081ⁿ	6500479	0.400

Table 5: Light viewer, user 2593914277176566399 (17 airings watched ≥ 80% in weeks 11-20, 13 distinct programmes).

tv_show_id	Title (from schedule)	Category	tv_show_id	Title (from schedule)	Category
20088	'Завтрак с 1+1'. Информационно-развлекательная программа.	Развлечения	5900347	'Деньги'.	Инфо
200338	Готовим вместе.	Познавательное	6500479	'Мир наизнанку'.	Познавательное
200344	'Чистоnews'.	Развлечения	12001682	М/с 'Губка Боб Квадратные Штаны', 89 с. 'Красти Губка'.	Детям
240081	Факты. День.	Инфо	12002896	М/с 'Вспыш и чудо-машинки', 4 сезон, 5 с. 'Суперогромный приз'.	Детям
400363	От пацанки до панянки, 4 сезон, 15 эп.	Развлечения	12002955	М/с 'Свинка Пеппа', 1 сезон, 11 с. 'Икота'.	Детям
400455	Любовь на выживание, 4 сезон, 5 эп.	Познавательное	90021962	Х/ф 'Эйс Вентура: Когда зовёт природа'.	Фильмы
2400467	'Викна-Новости'.	Инфо	90057599	Х/ф 'Ангелы Чарли 2: Только вперед'.	Фильмы
2400480	'ТСН. Неделя'.	Инфо	90072234	Х/ф 'Социальная сеть'.	Фильмы
2400508	Сегодня. Итоги с Олегом Панютой.	Инфо	90077275	Х/ф 'Миллион способов потерять голову'.	Фильмы
2500261	Подробности недели.	Инфо	90079862	Х/ф 'Глубоководный горизонт'.	Фильмы

Table 6: Titles and categories of every programme id in Tables 4-5, looked up from the schedule.

4.2 Kaggle submission

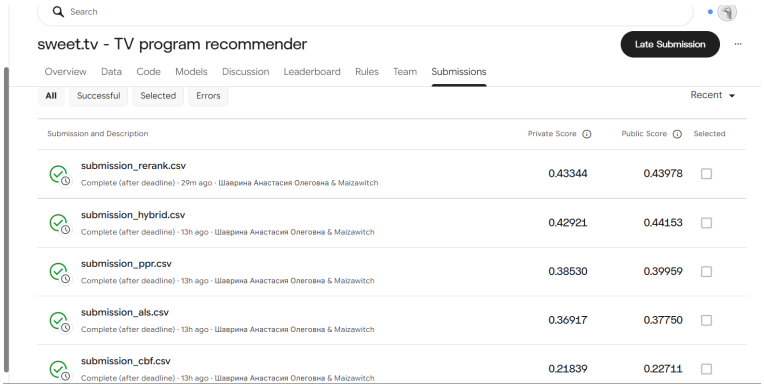


Figure 4: Kaggle Submissions page (late submissions, account Maizawitch). Designated final submission: submission_rerank.csv, returned private score 0.43344, public 0.43978. The popularity submission (private 0.28773) is below the visible area; all six scores are in Table 1 and results/kaggle_submissions.csv.

The designated final submission is `submissions/submission_rerank.csv` (ref 57803755), chosen on validation before its Kaggle score was known.

5. Discussion

5.1 Which recommendation strategy would we select for sweet.tv aand why?

Here we try to predict something that the user will WATCH, not something the user will LIKE. It's important to notice, because a person can watch something regularly as a habit. We can see on our examples that we are able to predict much better already watched content for an active user (AP@5=1.0), than new content for a less active one. Our models work better with a reoccuring data.

Let's also compare the methods: popularity already has a good result because a lot of users watch the same news and other kinds of regular shows. ALS tries to find more hidden similarities between users. PPR takes into account personal history really good (PPR 0.32 MAP@5 VS ALS 0.29 MAP@5). It happens because a random walk through the graph regularly returns to the programs of a particular user.

Content filtration is somehow worse than popularity: 0.18 VS 0.22. It is probably due to the fact that the similarity between genres and metadata does not indicate than a person will watch both because they can be on different channels and aat different time. People usually watch TV in the evening, so if the other show is in the morning, they won't see it.

The best signal here is channel exposure (the info which channels this user regularly watches). If we delete it, our hybrid model quality will drop significantly: 0.019 MAP@5. As I already said it is because people usually come after work and

turn on their favourite channels (same time, same channel).

That is why I've chosen a hybrid model with user history, popularity, ALS, PPR, content metadata and channel exposure + a LightGBM re-ranker on top. On validation my model gets MAP@5 =0.35, on Kaggle private 0.43. It is also cheap to compute at this scale: the whole test run takes about 3 minutes on an ordinary 8-core CPU.

But this method is not perfect because only 10.8% of the timetable has tv_show_id != 0. And also watching != being interested and != liking. So we predict that a person will watch something, not that they will like it.

5.2 Failure case, root cause, fix

The biggest technical problem was with the Personalized PageRank. At first, the validation was 452.8s, but it was too long for finding hyperparameters. So I changed it to 50 strongest connections in the graph, but MAP@5 dropped from 0.32 to 0.28. The problem was in the graph structure: it was almost dense, 71% pairs of the programs (among the 2,902 programs that had any viewer in weeks 11-20) shared a user. That's why we didn't get big advantage. And also for PageRank weak connection also have some useful info and we deleted it. In the end, I decided to use the whole graph, but add dense matrix float32 and optimized multiplication (BLAS). The MAP@5 didn't drop and the time became 141s.

There was also a small problem with LightGBM: the first reranking didn't work because of the dependancy on scikit-learn. I moved to API lgb.train.

5.3 What worked, what surprised me, next step

1) the validation is correct (80% threshold + check on the future weeks). The ranking of the algorithms is the same as the Kaggle private one.

2) those 89% of the data without tv_show_id where not thrown away: I used it for channel exposure. It was very helpful.

3) I was a bit surprised that content filtration is worse than popularity??? But then I started thinking about typical TV watcher and understood the reasons.

4) It is also interesting that ALS worked best with less factors and strong regularisation (30).

5) It is also interesting that all methods on Kaggle private have bettet scores on MAP@5. it is probably due to the longer history.

6) Due to the 5 we probably need to check how the length of the history affects our quality. And also we can add re-ranker candidates from the user's favourite channels.

6. AI Usage Disclosure

- **Tool:** Claude Code in the Claude desktop app, model Claude Opus 5.5 (`claude-opus-5-5`); one continuous session, exported unmodified as the attached session log.
- **How AI was used:** reading the exam brief, slides and course guidelines (docs); planning; writing all code in `src/` (code generation); running every experiment and Kaggle submission from my approval; debugging (PPR runtime, LightGBM wrapper error, memory blow-up in a feature); searching and checking references (search); figures (viz); drafting this report and the README.
- **What I personally verified:** **[TODO: list only what you actually did yourself, e.g. "hand-computed AP@5 for the heavy viewer's popularity list (0.643, matches Table 4)", "re-ran src/checks.py and compared the output with results/checks.json", "opened the 7 reference links", "checked the Kaggle scores on the Submissions page against Table 1"]**.
- **What I trusted without verification:** **[TODO: be honest, e.g. LightGBM's LambdaRank implementation, the correctness of the TF-IDF construction, the parquet conversion beyond the row counts]**.
- **Reference check:** every reference was checked against Crossref metadata or the publisher's page on 2026-10-30 (`notes/reference_check.md`); the 1999 PageRank technical report was dropped because its server returned an error.
- **Session log:** the attached session export is the audit trail for all of the above.

References

[1] ogurbych, "sweet.tv — TV program recommender," Kaggle community competition, 2020. [Online]. Available: https://www.kaggle.com/competitions/sweettv-tv-program-recommender. [Accessed: 2026-10-30].

[2] Y. Hu, Y. Koren, and C. Volinsky, "Collaborative filtering for implicit feedback datasets," in *Proc. 8th IEEE Int. Conf. Data Mining (ICDM)*, 2008, pp. 263-272, doi: 10.1109/ICDM.2008.22.

[3] T. H. Haveliwala, "Topic-sensitive PageRank," in *Proc. 11th Int. Conf. World Wide Web (WWW)*, 2002, pp. 517-526, doi: 10.1145/511446.511513.

[4] M. J. Pazzani and D. Billsus, "Content-based recommendation systems," in *The Adaptive Web*, LNCS, vol. 4321. Berlin, Germany: Springer, 2007, pp. 325-341, doi: 10.1007/978-3-540-72079-9_10.

[5] R. Agrawal, T. Imieliński, and A. Swami, "Mining association rules between sets of items in large databases," in *Proc. ACM SIGMOD Int. Conf. Management of Data*, 1993, pp. 207-216, doi: 10.1145/170035.170072.

[6] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, 2017. [Online]. Available: https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html. [Accessed: 2026-10-30].

[7] C. J. C. Burges, "From RankNet to LambdaRank to LambdaMART: An overview," Microsoft Research, Tech. Rep. MSR-TR-2010-82, Jun. 2010. [Online]. Available: https://www.microsoft.com/en-us/research/publication/from-ranknet-to-lambdarank-to-lambdamart-an-overview/. [Accessed: 2026-10-30].